

FPGA – Based Adaptive FIR Filter Using LMS Algorithm

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Abstract—This This paper describes the design and the testing of an adaptive FIR filter built on an FPGA using the Least Mean Squares (LMS) algorithm. The study compares different types of multipliers, such as the traditional array multiplier (conventional type), the Vedic multiplier, and the Binary Coded Decimal (BCD) multiplier, within a 3-tap FIR filter design. The multiplier unit has been the target for better architecture performance as it has a principal role in the finite impulse response (FIR) filter. Having been designed using Verilog, all conducted multiplier were tested to decide the best among them for the adaptive FIR filter. Testing the design was done in Quartus II and ModelSim software. Results illustrated that the BCD multiplier type was the most qualified through using only 12 registers and 151 logic elements compared to Array and Vedic multipliers. Obtained simulated results offers a maximum speed of 1282.05 MHz. Also, the performance of the design reflects an apparent error reducing of the tested signal with proper values compared to theoretically obtained results. Generally, the proposed FPGA based adaptive FIR system supports its effectiveness and reliability along with capability of high-speed operation.

Keywords—FPGA, Adaptive Filter, FIR Filter, BCD multiplier, LMS Algorithm.

I. INTRODUCTION

The Adaptive filter is of the most important filters in the field of digital systems. It derives its name from its ability to be accustomed to reach the desired values depending on various algorithm. Therefore, it is widely used in multiple applications such as signal processing, audio, control and communication systems. The idea of the adaptivity comes from the continuous system coefficients updating by performing number of iterations. Upon each iteration, new system (such as filter) coefficients are found. The optimum values of the coefficients is determined with respect to the difference between the tested signal and the desired one. This difference is defined as the error signal. Therefore, the adaptive filters contribute strongly in the systems where their input signal is changing over-time [1]. Recently, Field programmable Gate Arrays (FPGAs) have been considered as preferable for the implementation of digital signal processing systems (DSP) due to its rich arithmetic components such as registers and adders. Therefore, FPGA is regarded an ideal environment for implementing DSP circuits that require high performance [2]. For instance, implementing of adaptive Finite Input Response (FIR) filter through least mean square (LMS) algorithm is of the systems that needs high-performance devices to be designed. As a famous filter design,

FIR employs many adders, shifters and multipliers which turned it to be a good target for various and wide research. Of the most important and wide-used adaptive algorithms, LMS technique comes to be popular due to its obviousness in implementation. Therefore, it has been used in many practical applications [3],[4]. As a result, the implementing of FIR filters as an adaptive form through exploiting LMS technique, has become more effective when it is applied on FPGA platforms [5].

This paper presents the design of FIR filter based of LMS adaptive algorithm through FPGA. The BCD multiplier is involved in the design of the systems multipliers in order to enhance the performance of the filter. The performance is achieved through participating in reducing the digital components along with maximizing operational speed. The paper findings show that the proposed approach achieves the best minimizing of exploited digital elements as well as achieving superior processing 1speed.

In summary, this work provides a practical and scalable solution for high-performance, real-time signal processing applications using FPGA technology.

II. LITERATURE SURVEY

Various approaches have used FPGA software in order to design adaptive filters based on the LMS algorithm. For instance, an audio system processing exploited LMS in 3 different FPHGA designs for the aim of reducing the signal noise- ration (SNR) [6]. Also, in another approach, reduction of the power consumption was considered in [7] by using Veriglo HDL written architecture of an LMS-based adaptive filter system. In [2], several adaptive algorithms were designed and tested on FPGA platform. LMS algorithm were the most important tested algorithm since it reflects robust and simple calculations. The study in [2], created a system on chip (SOC) solution for various adaptive algorithms in terms of system speed, power consumption and resources of hardware. The LMS algorithm was coded in VHDL and implemented on FPGA.

Moreover, the study in [8] Cyclone IV EP4CE6 series FPGA was used to implement and adaptive-based LMS algorithm filter. The results of the ModelSim and the waveforms measured in the oscilloscope supported the possibility of using the approach in real time signal systems. Also, Using LMS and delay LMS (DLMS) algorithms has been done in the study of [9] where the region and capacity of the design are enhanced compared to the standard FIR

channel. The results were obtained by Modalism and a coded structure using Verilog HDL. Xilinx Vivado was used to compile the code. In [10], the property of FPGA flexibility and integrity was the reason behind choosing it to design LMS algorithm to be used in adaptive filter. Compared with digital signal processing-based filter design, the experimental results showed that the FPGA platform presents better accuracy and real-time adaptive filtering algorithm.

Finally, several studies have explored the broader applications of LMS-based adaptive filtering to improve signal processing. Adaptive filters have been shown to remove interference, predict relevant signals, and match the desired output effectively. Simulations indicate that the LMS algorithm converges quickly and reliably, allowing processed signals to closely follow expected values. Practical applications include correcting channel mismatches, enhancing system performance through adaptive equalizers, and filtering unwanted frequencies using adaptive notch filters. Overall, these studies confirm that LMS-based adaptive filters significantly enhance both signal quality and system reliability [11].

III. LMS ADAPTIVE FILTER ALGORITHM AND PROPOSED ARCHITECTURE

A. Adaptive LMS Algorithm

The LMS adaptive filter includes two main parts: a filtering section and an adaptation section. The filtering section employs a Finite Impulse Response (FIR) filter, whereas the adaptation section utilizes the LMS algorithm. During each iteration, the filtering section generates an output signal, which is then used to calculate an error value. This error is subsequently processed by the adaptation section to determine the weight adjustment term, which is used to update the weight vector for the following iteration [12]. Fig. 1 depicts a general adaptive filtering scheme.

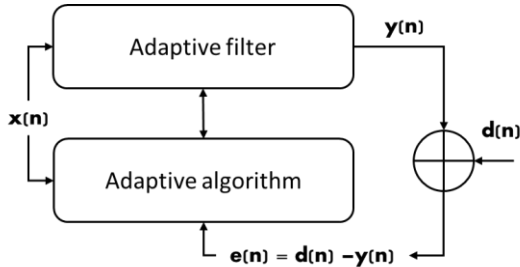


Fig. 1. General Adaptive Filter Scheme.

In an LMS adaptive filter, the weight vector $w(n)$ at the n^{th} iterations is updated for the aim of gaining the $(n+1)^{\text{th}}$ iteration through the following relation:

$$w(n+1) = w(n) + \mu x(n)e(n) \quad (1)$$

where μ represents the convergence factor.

The N -point weighted vector $w(n)$ can be defined as:

$$w(n) = [w_n(0), w_n(1), \dots, w_n(N-1), (0),]^T \quad (2)$$

The error signal $e(n)$ is calculated as:

$$e(n) = d(n) - y(n) \quad (3)$$

Here, $d(n)$ represents the desired response, while $y(n)$ is an estimate of this response, computed as:

$$y(n) = X^T(n)w(n) \quad (4)$$

The N -point input vector $x(n)$ is expressed as:

$$x(n) = [x(n), x(n-1), \dots, x(n-N+1), (0),]^T \quad (5)$$

B. Theoretical calculations

In order to test the reliability of our proposed design, the following example has been taken as a source during simulating processes. In this example, the input signal $x(n)$ was assumed to be $x(n) = [1 \ 2 \ 2 \ 5]$. The step-size parameter for LMS algorithm was suggested as $\mu = 1$. However, choosing this value is to present clearer results for the aim of comparison between theoretical and simulating results. Moreover, $h(n)$ (as the filter coefficients) is used to obtain the desired signal $d(n)$. It was assumed as $h(n) = [1 \ 2 \ 1]$. Therefore, the following results were taken in consideration for iteration 0, 1, and 2. Other iterations follow the same theoretical procedure. All calculations were performed depending on the equations 1, 2, 3, 4 and 5.

Through iteration 0,

$$x(0) = [x_0, x_{-1}, x_{-2}] = [1, 0, 0]$$

$$y(0) = w(0) \cdot x(0) = [0, 0, 0] \cdot [1, 0, 0] = 0$$

$$e(0) = d(0) - y(0) = 1 - 0 = 1$$

$$\Delta w = \mu \cdot e(0) \cdot x(0) = 1 \cdot 1 \cdot [1, 0, 0] = [1, 0, 0]$$

$$w(1) = w(0) + \Delta w = [0, 0, 0] + [1, 0, 0] = [1, 0, 0]$$

And through iteration 1,

$$x(1) = [x_1, x_0, x_{-1}] = [2, 1, 0]$$

$$y(1) = w(1) \cdot x(1) = [1, 0, 0] \cdot [2, 1, 0]$$

$$= 1 \cdot 2 + 0 \cdot 1 + 0 \cdot 0 = 2$$

$$e(1) = d(1) - y(1) = 4 - 2 = 2$$

$$\Delta w = 1 \cdot 2 \cdot [2, 1, 0] = [4, 2, 0]$$

$$w(2) = w(1) + \Delta w = [1, 0, 0] + [4, 2, 0] = [5 \ 2 \ 0]$$

IV. PROPOSED DESIGN CIRCUIT

A. FIR filter architecture

The architecture of the proposed adaptive filter design comprises an FIR filter circuit along with the necessary arithmetic components required to complete the overall system. The FIR filter circuit includes multipliers, which play a crucial role due to their significant impact on the output results, as well as registers and arithmetic adders. A schematic representation of the proposed 3-tap FIR filter circuit is shown in Fig. 2.

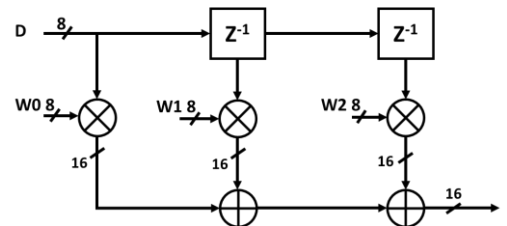


Fig. 2. A Block-circuit diagram of 3-tap FIR filter

FIR systems are crucial in DSPs since they experience linear phase characteristics and an innate stability. However, the overall performance of these filters is strongly influenced by the efficiency of the multiplier components within the filter architecture. In this research, the integration of various multiplier designs is investigated. In addition to the commonly used array multiplier, valued for its simplicity and regular structure [13], Vedic multipliers [14] and Binary Coded Decimal (BCD) multipliers [15] have also been incorporated into a standard FIR filter framework to assess

their impact on filter performance. Each multiplier was independently implemented using Verilog Hardware Description Language (HDL) and subsequently embedded within the FIR filter structure. The proposed 3-tap FIR filter design was coded and synthesized using Verilog HDL and Quartus II, and its gate-level behavior was simulated on the Cyclone II FPGA kit platform. The Register Transfer Level (RTL) view of the FIR filter incorporating the BCD-multiplier is shown in Fig. 3.

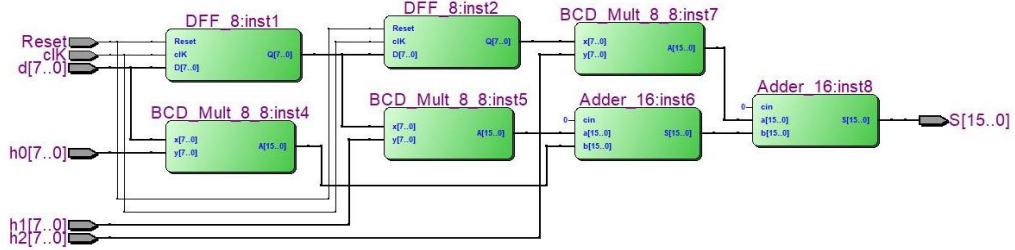


Fig. 3. RTL viewer of the proposed FIR filter incorporating the BCD-multiplier.

B. Adaptive FIR filter architecture

Unlike fixed-coefficient FIR filters, adaptive filters update their coefficients in real time to optimize performance, making them suitable for various signal processing applications such as systems of channel equalizer, noise cancelling and identification. The proposed adaptive FIR filter utilizes the Least Mean Squares (LMS) algorithm to adjust the filter coefficients. The main components of the system include the input signal generation, which can be simulated or real-time signals fed into the filter, and the adaptive filter core, which performs the FIR filtering operation with adjustable coefficients. Fig. 4 represents the general scheme for the suggested 3-tap FIR adaptive filtering system with LMS algorithm involved.

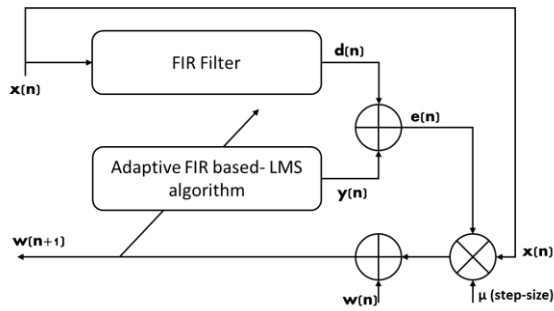


Fig. 4. General scheme for an FIR adaptive filter based on LMS algorithm.

The adaptive FIR filter was coded in Verilog HDL and synthesized through Quartus II. Simulation results were performed in ModelSim in order to prove the accuracy of the LMS algorithm. Those simulations refer to data inputs denoted as (D), coefficient values represented as (W_0 , W_1 and W_2), LMS-obtained coefficients (W_3 , W_4 and W_5), and the estimated signal output.

V. THE SIMULATION RESULTS

The proposed simulation results of the gate level show the data inputs (D), (W_0 , W_1 and W_2) as the coefficient values along with the expected output value denoted as (Y). Fig. 5 indicated that the calculated outputs are thoroughly matched with the theoretically calculated values. Table 1 illustrates the findings of the proposed multipliers in term of logic elements and speed of operation.

TABLE 1 COMPARATIVE RESULTS OF THE PROPOSED MULTIPLIERS

Multiplier	Array	BCD	Vedic
Max. operating speed (MHz)	1082.66	1067.29	1153.4
Logic Elements	497	151	390

Table 1 indicates that the multipliers exhibit comparable operating speeds, with the Vedic multiplier showing a slight advantage. In contrast, the BCD multiplier demonstrates significant efficiency in terms of logic element usage. As a result, the BCD multiplier was identified as the most optimal choice for the proposed FIR filter design.

For the adaptive LMS algorithm part, the simulations results present the data inputs (D), the LMS-based coefficient updates (W_3 , W_4 and W_5), the coefficient values of the desired signal (W_0 , W_1 and W_2) along with the estimated output as shown in Fig. 6. Verification and simulating of the proposed architecture were performed using Quartus II and ModelSim software. Obtained findings illustrates that the maximum operation speed is of 1282.05 Mhz with 137 logic elements and 16 register requirements.

The former results refer to the effective error minimizing of the proposed architecture. Moreover, the involving of the BCD multiplier into the design of the adaptive filter achieves not only performing enhancement but also participate mainly in system scalability.

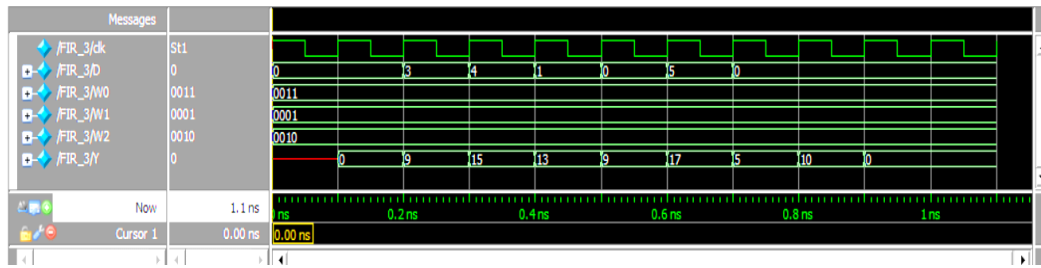


Fig. 5. Gate level simulation of the FIR filter.

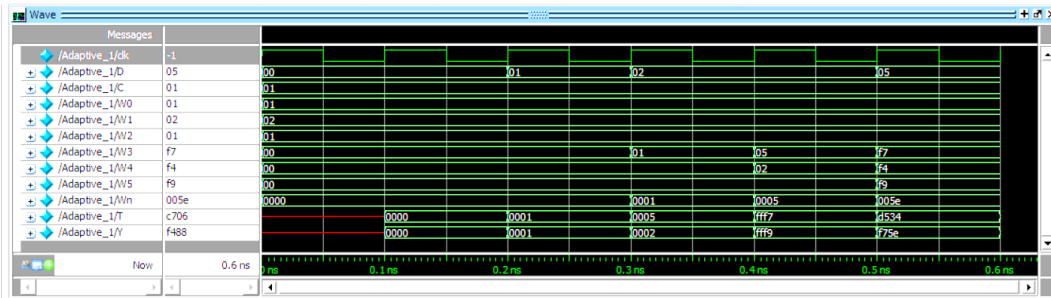


Fig. 6. The Simulation result of adaptive FIR filter.

VI. THE CONCLUSION

The conclusion drawn behind this paper illustrates that involving the BCD multiplier in the architecture of the FPGA-based adaptive FIR filter mainly improves the performance, efficiency, and design scalability. Recorded numbers through Quartus II and a ModelSim Software confirmed that the proposed filter runs at 151 logic elements and 12 registers along with a 1067.29 MHz as a maximum operation speed. Hence, this refers to highlighting the design's suitability for high-speed, resource-constrained FPGA environments. The comparison between theoretical and simulated results shows a clear agreement which validates the dependability and effectiveness of the methodology. Obtained results advocate that the FIR filters designed in FPGA are more accurate and robust when combined with an efficient multiplier such as the BCD multiplier. They can also preserve low resource consumption along with supporting complex real time system processing. For the future work, the proposed work could be developed through involving various techniques such as advance multiplier designs, multi-channel adaptive algorithm and pipeline optimization. Also, further exploration of power efficiency and scalability to higher-order FIR filters is acknowledged as future work.

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